

Question ID 3d1070c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3d1070c9

The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

- A. 50
- B. 57
- C. 80
- D. 110

ID: 3d1070c9 Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that the function f is defined by $f(x) = 25x + 30$. Substituting 2 for x in this equation yields $f(2) = 25(2) + 30$, which is equivalent to $f(2) = 50 + 30$, or $f(2) = 80$. Therefore, the value of $f(x)$ is 80 when $x = 2$.

Choice A is incorrect. This is the value of $25(2)$, not $25(2) + 30$.

Choice B is incorrect. This is the value of $25 + 2 + 30$, not $25(2) + 30$.

Choice D is incorrect. This is the value of $(25 + 30)(2)$, not $25(2) + 30$.

Question Difficulty:

Easy

Question ID bd9eb2b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bd9eb2b5

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

- A. 8
- B. 9
- C. 64
- D. 80

ID: bd9eb2b5 Answer

Correct Answer:

B

Rationale

Choice B is correct. Substituting 72 for $f(x)$ in the given function yields $72 = 8x$. Dividing each side of this equation by 8 yields $9 = x$. Therefore, $f(x) = 72$ when the value of x is 9 .

Choice A is incorrect. This is the value of x for which $f(x) = 64$, not $f(x) = 72$.

Choice C is incorrect. This is the value of x for which $f(x) = 512$, not $f(x) = 72$.

Choice D is incorrect. This is the value of x for which $f(x) = 640$, not $f(x) = 72$.

Question Difficulty:

Easy

Question ID 1480dd5c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1480dd5c

$$f(x) = 4x + b$$

For the linear function f , b is a constant and $f(7) = 28$. What is the value of b ?

- A. 0
- B. 1
- C. 4
- D. 7

ID: 1480dd5c Answer

Correct Answer:
A

Rationale

Choice A is correct. For the linear function f , it's given that $f(7) = 28$. Substituting 7 for x and 28 for $f(x)$ in the given function yields $28 = 4(7) + b$, or $28 = 28 + b$. Subtracting 28 from each side of this equation yields $0 = b$. Therefore, the value of b is 0 .

Choice B is incorrect. Substituting 1 for b in the given function yields $f(x) = 4x + 1$. For this function, when the value of x is 7 , the value of $f(x)$ is 29 , not 28 .

Choice C is incorrect. Substituting 4 for b in the given function yields $f(x) = 4x + 4$. For this function, when the value of x is 7 , the value of $f(x)$ is 32 , not 28 .

Choice D is incorrect. Substituting 7 for b in the given function yields $f(x) = 4x + 7$. For this function, when the value of x is 7 , the value of $f(x)$ is 35 , not 28 .

Question Difficulty:
Easy

Question ID 0d6ab461

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0d6ab461

Gabriella deposits **\$35** in a savings account at the end of each week. At the beginning of the **1st** week of a year there was **\$600** in that savings account. How much money, in dollars, will be in the account at the end of the **4th** week of that year?

- A. **460**
- B. **635**
- C. **639**
- D. **740**

ID: 0d6ab461 Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that at the beginning of the **1st** week of the year there was **\$600** in a savings account and Gabriella deposits **\$35** in that savings account at the end of each week. Therefore, the amount of money, in dollars, in the savings account at the end of the **4th** week of that year is **$600 + 4(35)$** , or **740**.

Choice A is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the **4th** week if Gabriella withdraws, rather than deposits, **\$35** at the end of each week.

Choice B is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the **1st** week, not the **4th** week.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 520c8177

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 520c8177

A veterinarian recommends that each day a certain rabbit should eat **25** calories per pound of the rabbit’s weight, plus an additional **11** calories. Which equation represents this situation, where ***c*** is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit’s weight is ***x*** pounds?

- A. $c = 25x$
- B. $c = 36x$
- C. $c = 11x + 25$
- D. $c = 25x + 11$

ID: 520c8177 Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that a veterinarian recommends that each day the rabbit should eat **25** calories per pound of the rabbit’s weight, plus an additional **11** calories. If the rabbit’s weight is ***x*** pounds, then multiplying **25** calories per pound by the rabbit’s weight, ***x*** pounds, yields **25*x*** calories. Adding the additional **11** calories that the rabbit should eat each day yields **25*x* + 11** calories. It’s given that ***c*** is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit’s weight is ***x*** pounds. Therefore, this situation can be represented by the equation **$c = 25x + 11$** .

Choice A is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **25** calories per pound of the rabbit’s weight.

Choice B is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **25 + 11**, or **36**, calories per pound of the rabbit’s weight.

Choice C is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **11** calories per pound of the rabbit’s weight, plus an additional **25** calories.

Question Difficulty:
Easy

Question ID 88e13c8c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 88e13c8c

The total cost $f(x)$, in dollars, to lease a car for **36** months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is **\$400**?

- A. \$13,400
- B. \$13,000
- C. \$15,400
- D. \$37,400

ID: 88e13c8c Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that $f(x)$ is the total cost, in dollars, to lease a car from this dealership with a monthly payment of x dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is **\$400** is represented by the value of $f(x)$ when $x = 400$. Substituting **400** for x in the equation $f(x) = 36x + 1,000$ yields $f(400) = 36(400) + 1,000$, or $f(400) = 15,400$. Thus, when the monthly payment is **\$400**, the total cost to lease a car is **\$15,400**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 3a3b95df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3a3b95df

$$d = 16 - \frac{x}{30}$$

The equation shown gives the estimated amount of diesel d , in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

- A. 0
- B. 6
- C. 14
- D. 16

ID: 3a3b95df Answer

Correct Answer:
B

Rationale

Choice B is correct. It’s given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel d , in gallons, that remains in the gas tank of the truck after being driven x miles. Substituting **300** for x in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is **6** gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

Question Difficulty:
Easy

Question ID 84664a7c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 84664a7c

The front of a roller-coaster car is at the bottom of a hill and is 15 feet above the ground. If the front of the roller-coaster car rises at a constant rate of 8 feet per second, which of the following equations gives the height h , in feet, of the front of the roller-coaster car s seconds after it starts up the hill?

- A. $h = 8s + 15$
- B. $h = 15s + \frac{335}{8}$
- C. $h = 8s + \frac{335}{15}$
- D. $h = 15s + 8$

ID: 84664a7c Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that the front of the roller-coaster car starts rising when it’s 15 feet above the ground. This initial height of 15 feet can be represented by a constant term, 15, in an equation. Each second, the front of the roller-coaster car rises 8 feet, which can be represented by $8s$. Thus, the equation $h = 8s + 15$ gives the height, in feet, of the front of the roller-coaster car s seconds after it starts up the hill.

Choices B and C are incorrect and may result from conceptual errors in creating a linear equation. Choice D is incorrect and may result from switching the rate at which the roller-coaster car rises with its initial height.

Question Difficulty:
Easy

Question ID 7fac16fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 7fac16fb

The function f is defined by $f(x) = \frac{7}{10}x + 55$. What is the value of $f(20)$?

ID: 7fac16fb Answer

Correct Answer:
69

Rationale

The correct answer is 69. The value of $f(20)$ can be found by evaluating the function $f(x) = \frac{7}{10}x + 55$ for $x = 20$. Substituting 20 for x in this function yields $f(20) = \frac{7}{10}(20) + 55$, or $f(20) = 69$. Therefore, the value of $f(20)$ is 69.

Question Difficulty:
Easy

Question ID 06fc1726

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 06fc1726

If f is the function defined by $f(x) = \frac{2x - 1}{3}$,
what is the value of $f(5)$?

- A. $\frac{4}{3}$
- B. $\frac{7}{3}$
- C. 3
- D. 9

ID: 06fc1726 Answer

Correct Answer:
C

Rationale

Choice C is correct. If $f(x) = \frac{2x - 1}{3}$, then $f(5) = \frac{2(5) - 1}{3} = \frac{10 - 1}{3} = \frac{9}{3} = 3$.

Choice A is incorrect and may result from not multiplying x by 2 in the numerator. Choice B is incorrect and may result from dividing 2x by 3 and then subtracting 1. Choice D is incorrect and may result from evaluating only the numerator 2x – 1.

Question Difficulty:
Easy

Question ID 6863c7ce

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 6863c7ce

$$d = 16t$$

The given equation represents the distance d , in inches, where t represents the number of seconds since an object started moving. Which of the following is the best interpretation of 16 in this context?

- A. The object moved a total of 16 inches.
- B. The object moved a total of $16t$ inches.
- C. The object is moving at a rate of 16 inches per second.
- D. The object is moving at a rate of $\frac{1}{16}$ inches per second.

ID: 6863c7ce Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that in the equation $d = 16t$, d represents the distance, in inches, and t represents the number of seconds since an object started moving. In this equation, t is being multiplied by 16 . This means that the object's distance increases by 16 inches each second. Therefore, the best interpretation of 16 in this context is that the object is moving at a rate of 16 inches per second.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is the best interpretation of $16t$, rather than 16 , in this context.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Easy

Question ID 0b332f00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0b332f00

The function g is defined by $g(x) = 6x$. For what value of x is $g(x) = 54$?

ID: 0b332f00 Answer

Correct Answer:
9

Rationale

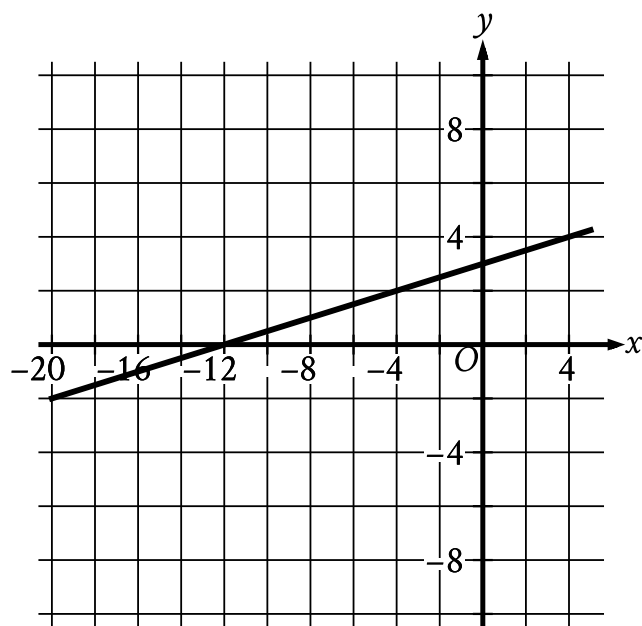
The correct answer is 9. It's given that $g(x) = 6x$. Substituting 54 for $g(x)$ in the given function yields $54 = 6x$. Dividing both sides of this equation by 6 yields $x = 9$. Therefore, the value of x when $g(x) = 54$ is 9.

Question Difficulty:
Easy

Question ID c10ad793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c10ad793



The graph of the linear function f is shown, where $y = f(x)$. What is the x -intercept of the graph of f ?

- A. $(-12, 0)$
- B. $(0, 0)$
- C. $(\frac{1}{4}, 0)$
- D. $(12, 0)$

ID: c10ad793 Answer

Correct Answer:

A

Rationale

Choice A is correct. The x -intercept of a graph is the point where the graph intersects the x -axis. The graph of function f , where $y = f(x)$, intersects the x -axis at $(-12, 0)$. Therefore, the x -intercept of the graph of f is $(-12, 0)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

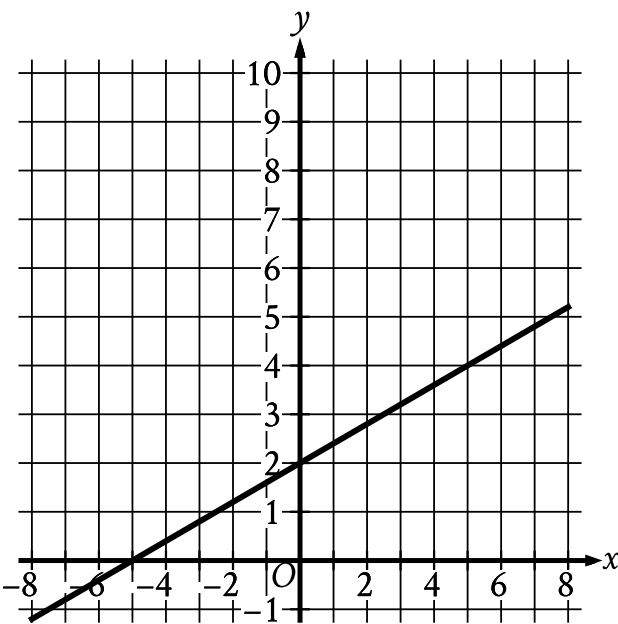
Question Difficulty:

Easy

Question ID d11910d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: d11910d6



The graph of the linear function f is shown. What is the y-intercept of the graph of $y = f(x)$?

- A. $(-5, 0)$
- B. $(2, 0)$
- C. $(0, 2)$
- D. $(0, -5)$

ID: d11910d6 Answer

Correct Answer:

C

Rationale

Choice C is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The graph of $y = f(x)$ shown intersects the y-axis at the point $(0, 2)$. Therefore, the y-intercept of the graph of $y = f(x)$ is $(0, 2)$.

Choice A is incorrect. This is the x-intercept, not the y-intercept, of the graph of $y = f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 0eae6be1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0eae6be1

The number y is ~~84~~ less than the number x . Which equation represents the relationship between x and y ?

- A. $y = x + 84$
- B. $y = \frac{1}{84}x$
- C. $y = 84x$
- D. $y = x - 84$

ID: 0eae6be1 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the number y is ~~84~~ less than the number x . A number that's ~~84~~ less than the number x is equivalent to ~~84~~ subtracted from the number x , or $x - 84$. Therefore, the equation $y = x - 84$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Easy

Question ID 361f97c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 361f97c7

The function f is defined by $f(x) = 4x - 3$. What is the value of $f(10)$?

- A. -30
- B. 37
- C. 40
- D. 43

ID: 361f97c7 Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 4x - 3$. Substituting 10 for x in the given function yields $f(10) = 4(10) - 3$, which is equivalent to $f(10) = 40 - 3$, or $f(10) = 37$. Therefore, the value of $f(10)$ is 37 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f(10)$ for the function $f(x) = 4x$, not $f(x) = 4x - 3$.

Choice D is incorrect. This is the value of $f(10)$ for the function $f(x) = 4x + 3$, not $f(x) = 4x - 3$.

Question Difficulty:

Easy

Question ID 447fa970

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 447fa970

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

ID: 447fa970 Answer

Correct Answer:
30

Rationale

The correct answer is 30. The value of $f(x)$ when $x = 4$ can be found by substituting 4 for x in the given equation $f(x) = 7x + 2$. This yields $f(4) = 7(4) + 2$, or $f(4) = 30$. Therefore, when $x = 4$, the value of $f(x)$ is 30.

Question Difficulty:
Easy

Question ID 27198699

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 27198699

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of **33** in this context?

- A. The estimated height, in cm, of the liquid at the start of the project
- B. The estimated height, in cm, of the liquid at the end of the project
- C. The estimated change in the height, in cm, of the liquid each day
- D. The estimated number of days for all of the liquid to evaporate

ID: 27198699 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that the function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. For a linear function in the form $f(x) = a + bx$, where a and b are constants, a represents the value of $f(0)$ and b represents the rate of change of the function. It follows that in the given function, **33** represents the value of $f(0)$. Therefore, the best interpretation of **33** in this context is the estimated height, in cm, of the liquid at the start of the project.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The estimated change in the height, in cm, of the liquid each day is **0.18**, not **33**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID a130fcdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a130fcdc

$g(x) = 11x + 4$

For the given linear function g , which table shows three values of x and their corresponding values of $g(x)$?

A.

x	$g(x)$
-1	7
0	11
1	15

B.

x	$g(x)$
-1	-4
0	0
1	4

C.

x	$g(x)$
-1	-7
0	4
1	15

D.

x	$g(x)$
-1	-11
0	0
1	11

ID: a130fcdc Answer

Correct Answer:

C

Rationale

Choice C is correct. Each of the tables shows the same three values of x : -1 , 0 , and 1 . Substituting -1 for x in the given function yields $g(-1) = 11(-1) + 4$, or $g(-1) = -7$. Therefore, when $x = -1$, the corresponding value of $g(x)$ is -7 . Substituting 0 for x in the given function yields $g(0) = 11(0) + 4$, or $g(0) = 4$. Therefore, when $x = 0$, the corresponding value of $g(x)$ is 4 . Substituting 1 for x in the given function yields $g(1) = 11(1) + 4$, or $g(1) = 15$. Therefore, when $x = 1$, the corresponding value of $g(x)$ is 15 . The table in choice C shows -7 , 4 , and 15 as the corresponding value of $g(x)$ for x -values of -1 , 0 , and 1 , respectively. Therefore, the table in choice C shows three values of x and their corresponding values of $g(x)$.

Choice A is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x + 11$.

Choice B is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x$.

Choice D is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 11x$.

Question Difficulty:

Easy

Question ID bf36c815

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bf36c815

The function g is defined by $g(x) = -x + 8$.

What is the value of $g(0)$?

- A. -8
- B. 0
- C. 4
- D. 8

ID: bf36c815 Answer

Correct Answer:

D

Rationale

Choice D is correct. The value of $g(0)$ is found by substituting 0 for x in the function g . This yields $g(0) = -0 + 8$, which can be rewritten as $g(0) = 8$.

Choice A is incorrect and may result from misinterpreting the equation as $g(x) = x + (-8)$ instead of $g(x) = -x + 8$. Choice B is incorrect. This is the value of x , not $g(x)$. Choice C is incorrect and may result from calculation errors.

Question Difficulty:

Easy

Question ID 0d391910

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0d391910

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

ID: 0d391910 Answer

Correct Answer:

2

Rationale

The correct answer is **2**. Substituting **8** for $f(x)$ in the given equation yields $8 = 4x$. Dividing the left- and right-hand sides of this equation by **4** yields $x = 2$. Therefore, the value of x is **2** when $f(x) = 8$.

Question Difficulty:

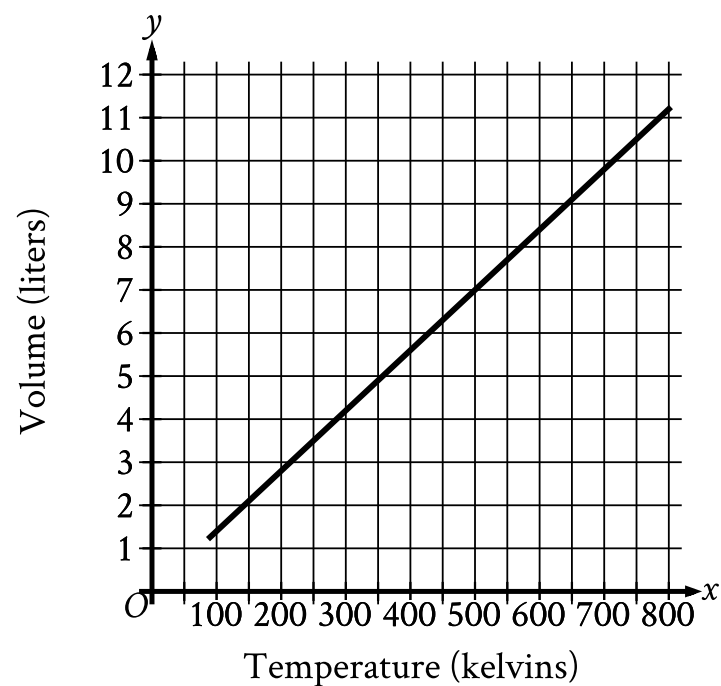
Easy

Question ID 930c2990

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 930c2990

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y , in liters, of the hydrogen when its temperature is x kelvins.



What is the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins?

- A. 0
- B. $\frac{7}{500}$
- C. 7
- D. $\frac{500}{7}$

ID: 930c2990 Answer

Correct Answer:
C

Rationale

Choice C is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents volume, in liters. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins is represented by the y -coordinate of the point on the graph that has an x -coordinate of **500**. The point on the graph with an x -coordinate of **500** has a y -coordinate of **7**. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins is **7**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:

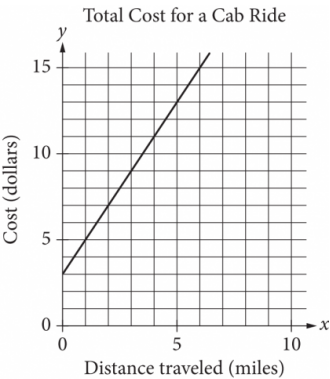
Easy

Question ID 3f5375d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3f5375d9

The line graphed in the xy -plane below models the total cost, in dollars, for a cab ride, y , in a certain city during nonpeak hours based on the number of miles traveled, x .



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

- A. \$2.00
- B. \$2.60
- C. \$3.00
- D. \$5.00

ID: 3f5375d9 Answer

Correct Answer:
A

Rationale

Choice A is correct. The cost of each additional mile traveled is represented by the slope of the given line. The slope of the line can be calculated by identifying two points on the line and then calculating the ratio of the change in y to the change in x between the two points. Using the points $(1, 5)$ and $(2, 7)$, the slope is equal to $\frac{7-5}{2-1}$, or 2. Therefore, the cost for each additional mile traveled of the cab ride is \$2.00.

Choice B is incorrect and may result from calculating the slope of the line that passes through the points $(5, 13)$ and $(0, 0)$. However, $(0, 0)$ does not lie on the line shown. Choice C is incorrect. This is the y -coordinate of the y -intercept of the graph and represents the flat fee for a cab ride before the charge for any miles traveled is added. Choice D is incorrect. This value represents the total cost of a 1-mile cab ride.

Question Difficulty:
Easy

Question ID 1d18794b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1d18794b

A contract for a certain service requires a onetime activation cost of **\$35** and a monthly cost of **\$23**. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

- A. $c = \frac{t}{23} + 35$
- B. $c = \frac{t}{35} + 23$
- C. $c = 23t + 35$
- D. $c = 35t + 23$

ID: 1d18794b Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that this service contract requires a monthly cost of **\$23**. A monthly cost of **\$23** for t months results in a cost of **$23t$** . It's also given that this service contract requires a onetime activation cost of **\$35**. Adding the onetime activation cost to the monthly cost of the service contract for t months yields the total cost c , in dollars, of this service contract for t months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID b51c173d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: b51c173d

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 2 and has a y -intercept at $(0, -5)$. Which equation defines f ?

- A. $f(x) = \frac{1}{2}x - 5$
- B. $f(x) = -\frac{1}{2}x - 5$
- C. $f(x) = -2x - 5$
- D. $f(x) = 2x - 5$

ID: b51c173d Answer

Correct Answer:
D

Rationale

Choice D is correct. An equation defining the linear function f can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. It's given that the graph of $y = f(x)$ has a slope of 2 . Therefore, $m = 2$. It's also given that the graph of $y = f(x)$ has a y -intercept at $(0, -5)$. Therefore, $b = -5$. Substituting 2 for m and -5 for b in the equation $f(x) = mx + b$ yields $f(x) = 2x - 5$. Thus, the equation that defines f is $f(x) = 2x - 5$.

Choice A is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{2}$, not 2 .

Choice B is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane has a slope of $-\frac{1}{2}$, not 2 .

Choice C is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane has a slope of -2 , not 2 .

Question Difficulty:
Easy

Question ID 4702da8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4702da8f

The function f is defined by $f(x) = 80 - 6x$. What is the value of $f(7)$?

- A. 13
- B. 38
- C. 74
- D. 81

ID: 4702da8f Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that function f is defined by $f(x) = 80 - 6x$. The value of $f(7)$ can be found by substituting 7 for x in the given function, which yields $f(7) = 80 - 6(7)$, or $f(7) = 80 - 42$, which is equivalent to $f(7) = 38$. Therefore, the value of $f(7)$ is 38.

Choice A is incorrect. This is the value of $80 - 67$, not $80 - 6(7)$.

Choice C is incorrect. This is the value of $80 - 6(1)$, not $80 - 6(7)$.

Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 6(7)$.

Question Difficulty:

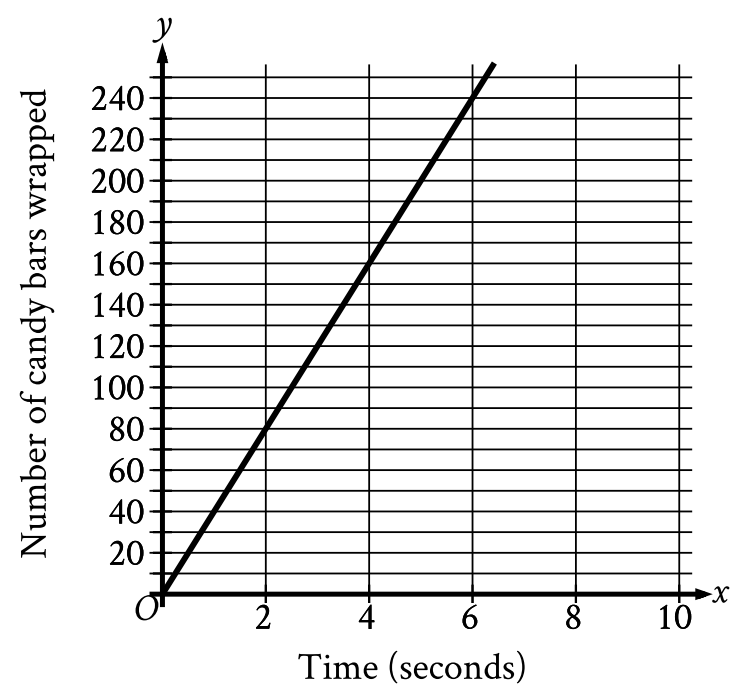
Easy

Question ID 13294295

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 13294295

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

- A. 2
- B. 40
- C. 78
- D. 80

ID: 13294295 Answer

Correct Answer:

B

Rationale

Choice B is correct. For the graph shown, the x -axis represents time, in seconds, and the y -axis represents the number of candy bars wrapped. The slope of a line in the xy -plane is the change in y for each 1-unit increase in x . It follows that the slope of the graph shown represents the estimated number of candy bars the machine wraps with a label per second. The slope, m , of a line in the xy -plane can be found using any two points, (x_1, y_1) and (x_2, y_2) , on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shown passes through the points $(0, 0)$ and $(2, 80)$. Substituting these points for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{80 - 0}{2 - 0}$, which is equivalent to $m = \frac{80}{2}$, or $m = 40$. Therefore, the estimated number of candy bars the machine wraps with a label per second is 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

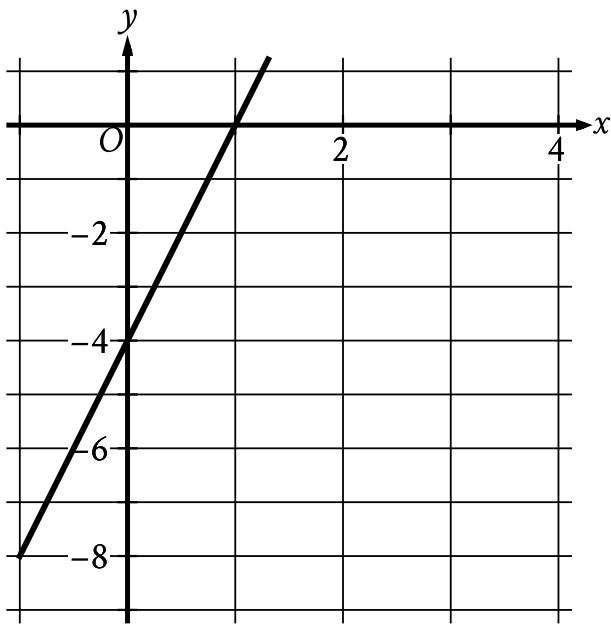
Question Difficulty:

Easy

Question ID 33e4af6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 33e4af6b



The graph of the function f is shown, where $y = f(x)$. What is the y -intercept of the graph?

- A. $(0, -1)$
- B. $(0, -4)$
- C. $(0, 1)$
- D. $(0, 4)$

ID: 33e4af6b Answer

Correct Answer:

B

Rationale

Choice B is correct. The y -intercept of a graph is the point where the graph intersects the y -axis. The graph of function f shown intersects the y -axis at the point $(0, -4)$. Therefore, the y -intercept of the graph is $(0, -4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 7b17f86a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 7b17f86a

$$f(x) = x + \frac{8}{11}$$

The function f is defined by the given equation. What is the value of $f(x)$ when $x = \frac{3}{11}$?

ID: 7b17f86a Answer

Correct Answer:

1

Rationale

The correct answer is **1**. It's given that the function f is defined by $f(x) = x + \frac{8}{11}$. Substituting $\frac{3}{11}$ for x in the given function yields $f(\frac{3}{11}) = \frac{3}{11} + \frac{8}{11}$, which gives $f(\frac{3}{11}) = \frac{11}{11}$, or $f(\frac{3}{11}) = 1$. Therefore, when $x = \frac{3}{11}$, the value of $f(x)$ is **1**.

Question Difficulty:

Easy

Question ID adf60b28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: adf60b28

$$f(x) = 8x + 4$$

The function f gives the estimated height, in feet, of a willow tree x years after its height was first measured. Which statement is the best interpretation of 4 in this context?

- A. The tree will be measured each year for 4 years.
- B. The tree is estimated to grow to a maximum height of 4 feet.
- C. The estimated height of the tree increased by 4 feet each year.
- D. The estimated height of the tree was 4 feet when it was first measured.

ID: adf60b28 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the function $f(x) = 8x + 4$ gives the estimated height, in feet, of a willow tree x years after its height was first measured. For a function defined by an equation of the form $f(x) = mx + b$, where m and b are constants, b represents the value of $f(x)$ when $x = 0$. It follows that in the given function, 4 represents the value of $f(x)$ when $x = 0$. Therefore, the best interpretation of 4 in this context is that the estimated height of the tree was 4 feet when it was first measured.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Easy

Question ID b8cbe394

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: b8cbe394

Sean rents a tent at a cost of **\$11** per day plus a onetime insurance fee of **\$10**. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

- A. $c = 11(d + 10)$
- B. $c = 10(d + 11)$
- C. $c = 11d + 10$
- D. $c = 10d + 11$

ID: b8cbe394 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that the cost of renting a tent is **\$11** per day for d days. Multiplying the rental cost by the number of days yields **\$11 d** , which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of **\$10** to the rental cost of **\$11 d** gives the total cost c , in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was **\$($d + 11$)** for **10** days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was **\$10** and the onetime fee was **\$11**.

Question Difficulty:
Easy

Question ID 12983c1e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 12983c1e

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above.
Which of the following defines f ?

- A. $f(x) = 2x + 3$
- B. $f(x) = 3x + 2$
- C. $f(x) = 4x + 1$
- D. $f(x) = 5x$

ID: 12983c1e Answer

Correct Answer:
C

Rationale

Choice C is correct. Because f is a linear function of x , the equation $f(x) = mx + b$, where m and b are constants, can be used to define the relationship between x and $f(x)$. In this equation, m represents the increase in the value of $f(x)$ for every increase in the value of x by 1. From the table, it can be determined that the value of $f(x)$ increases by 8 for every increase in the value of x by 2. In other words, for the function f the value of m is $\frac{8}{2}$, or 4. The value of b can be found by substituting the values of x and $f(x)$ from any row of the table and the value of m into the equation $f(x) = mx + b$ and solving for b . For example, using $x = 1$, $f(x) = 5$, and $m = 4$ yields $5 = 4(1) + b$. Solving for b yields $b = 1$. Therefore, the equation defining the function f can be written in the form $f(x) = 4x + 1$.

Choices A, B, and D are incorrect. Any equation defining the linear function f must give values of $f(x)$ for corresponding values of x , as shown in each row of the table. According to the table, if $x = 3$, $f(x) = 13$. However, substituting $x = 3$ into the equation given in choice A gives $f(3) = 2(3) + 3$, or $f(3) = 9$, not 13. Similarly, substituting $x = 3$ into the equation given in choice B gives $f(3) = 3(3) + 2$, or $f(3) = 11$, not 13.

Lastly, substituting $x = 3$ into the equation given in choice D gives $f(3) = 5(3)$, or $f(3) = 15$, not 13. Therefore, the equations in choices A, B, and D cannot define f .

Question Difficulty:
Easy

Question ID aeaba0b6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: aeaba0b6

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of \$58?

ID: aeaba0b6 Answer

Correct Answer:
11

Rationale

The correct answer is 11. It's given that the function $f(x) = 14 + 4x$ represents the total cost, in dollars, of attending an arcade when x games are played. Substituting 58 for $f(x)$ in the given equation yields $58 = 14 + 4x$. Subtracting 14 from each side of this equation yields $44 = 4x$. Dividing each side of this equation by 4 yields $11 = x$. Therefore, 11 games can be played for a total cost of \$58.

Question Difficulty:
Easy

Question ID 70d9516e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 70d9516e

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d , in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be **2** seconds after passing the marker?

- A. **30**
- B. **32**
- C. **60**
- D. **90**

ID: 70d9516e Answer

Correct Answer:

C

Rationale

Choice C is correct. It’s given that t represents the number of seconds after the bus passes the marker. Substituting **2** for t in the given equation $d = 30t$ yields $d = 30(2)$, or $d = 60$. Therefore, the bus will be **60** feet from the marker **2** seconds after passing it.

Choice A is incorrect. This is the distance, in feet, the bus will be from the marker **1** second, not **2** seconds, after passing it.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the distance, in feet, the bus will be from the marker **3** seconds, not **2** seconds, after passing it.

Question Difficulty:

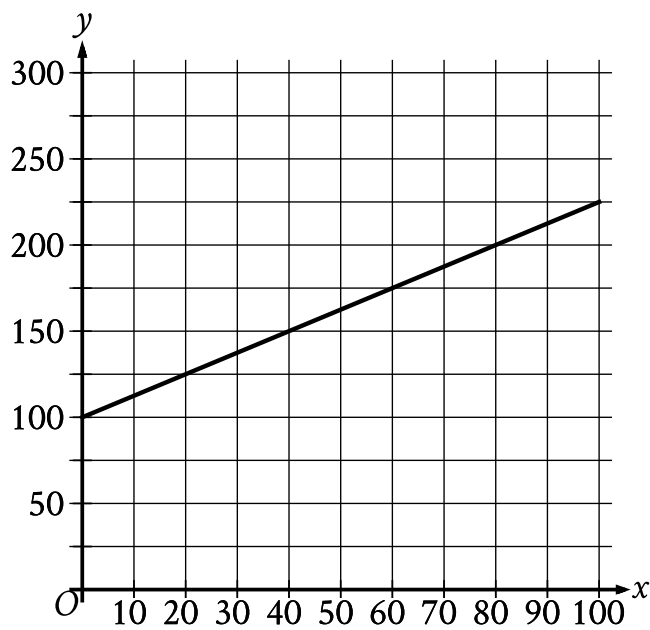
Easy

Question ID 720e51ac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 720e51ac

The cost y , in dollars, for a manufacturer to make x rings is represented by the line shown.



What is the cost, in dollars, for the manufacturer to make **60** rings?

- A. 100
- B. 125
- C. 175
- D. 225

ID: 720e51ac Answer

Correct Answer:
C

Rationale

Choice C is correct. The line shown represents the cost y , in dollars, for a manufacturer to make x rings. For the line shown, the x -axis represents the number of rings made by the manufacturer and the y -axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make **60** rings is represented by the y -coordinate of the point on the line that has an x -coordinate of **60**. The point on the line with an x -coordinate of **60** has a y -coordinate of **175**. Therefore, the cost, in dollars, for the manufacturer to make **60** rings is **175**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make **20** rings.

Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make **100** rings.

Question Difficulty:

Easy

Question ID 4e97f862

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4e97f862

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(7)$?

- A. 29
- B. 13
- C. -5
- D. -29

ID: 4e97f862 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 3x - 8$. The value of $f(7)$ is the value of $f(x)$ when $x = 7$. Substituting 7 for x in the given equation yields $f(7) = 3(7) - 8$, which is equivalent to $f(7) = 21 - 8$, or $f(7) = 13$.

Choice A is incorrect. This is the value of $f(7)$ when $f(x) = 3x + 8$, rather than $f(x) = 3x - 8$.

Choice C is incorrect. This is the value of $f(1)$, rather than $f(7)$.

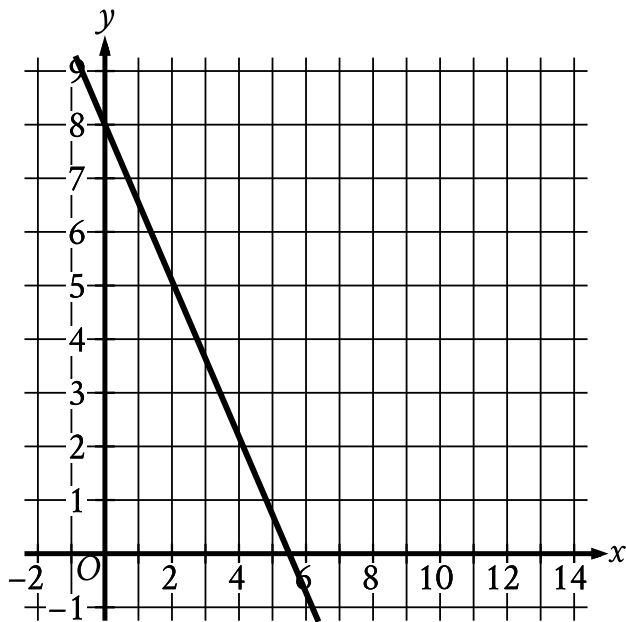
Choice D is incorrect. This is the value of $f(-7)$, rather than $f(7)$.

Question Difficulty:
Easy

Question ID 3174f07d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3174f07d



The graph of the linear function f is shown, where $y = f(x)$. What is the y-intercept of the graph of f ?

- A. $(0, 0)$
- B. $(0, -\frac{16}{11})$
- C. $(0, -8)$
- D. $(0, 8)$

ID: 3174f07d Answer

Correct Answer:
D

Rationale

Choice D is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The graph of function f shown intersects the y-axis at the point $(0, 8)$. Therefore, the y-intercept of the graph of f is $(0, 8)$.

Choice A is incorrect. This is the point where the x-axis, not the graph of f , intersects the y-axis.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID f79ffba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: f79ffba

The function h is defined by $h(x) = 3x - 7$. What is the value of $h(-2)$?

- A. -13
- B. -10
- C. 10
- D. 13

ID: f79ffba Answer

Correct Answer:

A

Rationale

Choice A is correct. The value of $h(-2)$ can be found by substituting -2 for x in the equation defining h . Substituting -2 for x in $h(x) = 3x - 7$ yields $h(-2) = 3(-2) - 7$, or $h(-2) = -13$. Therefore, the value of $h(-2)$ is -13 .

Choice B is incorrect. This is the value of $h(-1)$, not $h(-2)$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID a9039591

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a9039591

x	$f(x)$
0	29
1	32
2	35

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

- A. $f(x) = 3x + 29$
- B. $f(x) = 29x + 32$
- C. $f(x) = 35x + 29$
- D. $f(x) = 32x + 35$

ID: a9039591 Answer

Correct Answer:

A

Rationale

Choice A is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m and b are constants. It's given in the table that when $x = 0$, $f(x) = 29$. Substituting 0 for x and 29 for $f(x)$ in the equation $f(x) = mx + b$ yields $29 = m(0) + b$, or $29 = b$. Substituting 29 for b in the equation $f(x) = mx + b$ yields $f(x) = mx + 29$. It's also given in the table that when $x = 1$, $f(x) = 32$. Substituting 1 for x and 32 for $f(x)$ in the equation $f(x) = mx + 29$ yields $32 = m(1) + 29$, or $32 = m + 29$. Subtracting 29 from both sides of this equation yields $3 = m$. Substituting 3 for m in the equation $f(x) = mx + 29$ yields $f(x) = 3x + 29$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID a396ed75

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a396ed75

For a training program, Juan rides his bike at an average rate of **5.7** minutes per mile. Which function *m* models the number of minutes it will take Juan to ride *x* miles at this rate?

- A. $m(x) = \frac{x}{5.7}$
- B. $m(x) = x + 5.7$
- C. $m(x) = x - 5.7$
- D. $m(x) = 5.7x$

ID: a396ed75 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that Juan rides his bike at an average rate of **5.7** minutes per mile. The number of minutes it will take Juan to ride *x* miles can be determined by multiplying his average rate by the number of miles, *x*, which yields **5.7x**. Therefore, the function $m(x) = 5.7x$ models the number of minutes it will take Juan to ride *x* miles.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Easy

Question ID e4a588ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: e4a588ca

The function f is defined by $f(x) = \frac{1}{2}(x + 6)$. What is the value of $f(4)$?

- A. 20
- B. 12
- C. 10
- D. 5

ID: e4a588ca Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that the function f is defined by $f(x) = \frac{1}{2}(x + 6)$. Substituting 4 for x in the given function yields $f(4) = \frac{1}{2}(4 + 6)$, or $f(4) = 5$. Therefore, the value of $f(4)$ is 5 .

Choice A is incorrect. This is the value of $2(4 + 6)$, not $\frac{1}{2}(4 + 6)$.

Choice B is incorrect. This is the value of $2 + (4 + 6)$, not $\frac{1}{2}(4 + 6)$.

Choice C is incorrect. This is the value of $4 + 6$, not $\frac{1}{2}(4 + 6)$.

Question Difficulty:

Easy

Question ID 2e379126

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2e379126

The function g is defined by $g(x) = 4x - 6$. What is the value of $g(-7)$?

- A. ~~-34~~
- B. ~~-22~~
- C. ~~$-\frac{13}{4}$~~
- D. ~~$-\frac{1}{4}$~~

ID: 2e379126 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that the function g is defined by $g(x) = 4x - 6$. Substituting -7 for x into the given equation yields $g(-7) = 4(-7) - 6$, or $g(-7) = -34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $g(x) = -7$, not the value of $g(-7)$.

Question Difficulty:

Easy

Question ID 979b0b8d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 979b0b8d

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of **39** and passes through the point $(0, 0)$. Which equation defines f ?

- A. $f(x) = -39x$
- B. $f(x) = \frac{1}{39}x$
- C. $f(x) = x - 39$
- D. $f(x) = 39x$

ID: 979b0b8d Answer

Correct Answer:
D

Rationale

Choice D is correct. An equation defining a linear function can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. It's given that the graph of $y = f(x)$ has a slope of **39**, so $m = 39$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 0)$, so $b = 0$. Substituting **39** for m and **0** for b in $f(x) = mx + b$ yields $f(x) = 39x + 0$, or $f(x) = 39x$. Thus, the equation that defines f is $f(x) = 39x$.

Choice A is incorrect. This equation defines a function whose graph has a slope of **-39**, not **39**.

Choice B is incorrect. This equation defines a function whose graph has a slope of $\frac{1}{39}$, not **39**.

Choice C is incorrect. This equation defines a function whose graph has a slope of **1**, not **39**, and passes through the point $(0, -39)$, not $(0, 0)$.

Question Difficulty:
Easy

Question ID bf883fde

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bf883fde

For the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 and passes through the point $(0, -8)$. Which equation defines f ?

- A. $f(x) = 3x$
- B. $f(x) = 3x - 8$
- C. $f(x) = 3x + 5$
- D. $f(x) = 3x + 11$

ID: bf883fde Answer

Correct Answer:
B

Rationale

Choice B is correct. An equation defining a linear function can be written in the form $f(x) = mx + b$, where m and b are constants, m is the slope of the graph of $y = f(x)$ in the xy -plane, and $(0, b)$ is the y -intercept of the graph. It's given that for the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 . Therefore, $m = 3$. It's also given that this graph passes through the point $(0, -8)$. Therefore, the y -intercept of the graph is $(0, -8)$, and it follows that $b = -8$. Substituting 3 for m and -8 for b in the equation $f(x) = mx + b$ yields $f(x) = 3x - 8$. Thus, the equation that defines f is $f(x) = 3x - 8$.

Choice A is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 0)$, not $(0, -8)$.

Choice C is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$, not $(0, -8)$.

Choice D is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 11)$, not $(0, -8)$.

Question Difficulty:
Easy

Question ID 3462d850

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3462d850

Marisol drove 3 hours from City A to City B. The equation below estimates the distance d , in miles, Marisol traveled after driving for t hours.

$d = 45t$

Which of the following does 45 represent in the equation?

- A. Marisol took 45 trips from City A to City B.
- B. The distance between City A and City B is 45 miles.
- C. Marisol drove at an average speed of about 45 miles per hour.
- D. It took Marisol 45 hours to drive from City A to City B.

ID: 3462d850 Answer

Correct Answer:

C

Rationale

Choice C is correct. It’s given that d is the distance, in miles, Marisol traveled after driving for t hours. Therefore, 45 represents the distance in miles traveled per hour, which is the speed she drove in miles per hour.

Choice A is incorrect and may result from misidentifying speed as the number of trips. Choice B is incorrect and may result from misidentifying speed as the total distance. Choice D is incorrect and may result from misidentifying the speed as the time, in hours.

Question Difficulty:

Easy

Question ID fe6f9678

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: fe6f9678

For the linear function f , $f(0) = 17$ and $f(1) = 17$. Which equation defines f ?

- A. $f(x) = \frac{1}{17}$
- B. $f(x) = 1$
- C. $f(x) = 17$
- D. $f(x) = 34$

ID: fe6f9678 Answer

Correct Answer:
C

Rationale

Choice C is correct. An equation defining the linear function f can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. The slope of the graph of $y = f(x)$ can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are any two points that the graph passes through. If $f(0) = 17$, it follows that the graph of $y = f(x)$ passes through the point $(0, 17)$. If $f(1) = 17$, it follows that the graph of $y = f(x)$ passes through the point $(1, 17)$. Substituting $(0, 17)$ and $(1, 17)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{17 - 17}{1 - 0}$, which is equivalent to $m = \frac{0}{1}$, or $m = 0$. Since the graph of $y = f(x)$ passes through $(0, 17)$, it follows that $b = 17$. Substituting 0 for m and 17 for b in the equation $f(x) = mx + b$ yields $f(x) = 0x + 17$, or $f(x) = 17$. Thus, the equation that defines f is $f(x) = 17$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID bc638f2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bc638f2d

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. Which of the following is the best interpretation of 9 in this context?

- A. Tavon will keep the vine plant for 9 months.
- B. The vine plant is expected to grow 9 inches each month.
- C. The vine plant is expected to grow to a maximum length of 9 inches.
- D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

ID: bc638f2d Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. For a function defined by an equation of the form $f(t) = mt + b$, where m and b are constants, b represents the value of $f(0)$, or the value of $f(t)$ when the value of t is 0 . Therefore, for the function defined by $f(t) = 14t + 9$, 9 represents the value of $f(t)$ when the value of t is 0 . This means that 0 months after the vine plant was purchased, the estimated length of the vine plant was 9 inches. Therefore, the best interpretation of 9 in this context is the estimated length of the vine plant was 9 inches when Tavon purchased it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. The vine plant is expected to grow 14 inches, not 9 inches, each month.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 9d9fe1e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 9d9fe1e6

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, **in centimeters (cm)**, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of **39** in this context?

- A. The estimated height, **in cm**, of the fluid at the start of the experiment
- B. The estimated height, **in cm**, of the fluid at the end of the experiment
- C. The estimated change in the height, **in cm**, of the fluid each day
- D. The estimated number of days for all the fluid to evaporate

ID: 9d9fe1e6 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that the function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters (cm), of the fluid in the beaker x days after the start of the experiment. For a function defined by an equation of the form $f(x) = b + mx$, where m and b are constants, b represents the value of $f(x)$ when $x = 0$. It follows that in the given function, **39** represents the value of $f(x)$ when $x = 0$. Since $x = 0$ represents the start of the experiment, then the best interpretation of **39** in this context is that the estimated height, in cm, of the fluid was **39** at the start of the experiment.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. The estimated change in the height, in cm, of the fluid each day is **0.18**, not **39**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:

Easy

Question ID c4d49134

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c4d49134

$$s = 40 + 3t$$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

- A. 40
- B. 43
- C. 45
- D. 55

ID: c4d49134 Answer

Correct Answer:

D

Rationale

Choice D is correct. In the given equation, s is the speed, in miles per hour, of a certain car t seconds after it began to accelerate. Therefore, the speed of the car, in miles per hour, 5 seconds after it began to accelerate can be found by substituting 5 for t in the given equation, which yields $s = 40 + 3(5)$, or $s = 55$. Thus, the speed of the car 5 seconds after it began to accelerate is 55 miles per hour.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 255996a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 255996a6

$T = 1,000 + 18h$

In the equation above, T represents Brittany’s total take-home pay, in dollars, for her first week of work, where h represents the number of hours she worked that week and 1,000 represents a sign-on bonus. If Brittany’s total take-home pay was \$1,576, for how many hours was Brittany paid for her first week of work?

- A. 16
- B. 32
- C. 55
- D. 88

ID: 255996a6 Answer

Correct Answer:
B

Rationale

Choice B is correct. Since Brittany’s total take-home pay was \$1,576, the value 1,576 can be substituted for T in the given equation $T = 1,000 + 18h$ to give $1,576 = 1,000 + 18h$. Subtracting 1,000 from both sides of this equation gives $576 = 18h$. Dividing both sides of this equation by 18 gives $32 = h$. Therefore, Brittany was paid for 32 hours for her first week of work.

Choice A is incorrect. This is half the number of hours Brittany was paid for. Choice C is incorrect and may result from dividing 1,000 by 18. Choice D is incorrect and may result from dividing 1,576 by 18.

Question Difficulty:
Easy

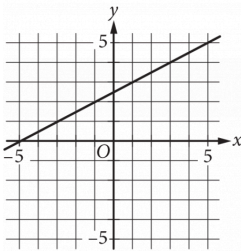
Question ID a8e6bd75

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

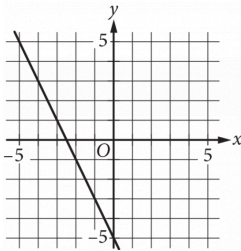
ID: a8e6bd75

Which of the following is the graph of the equation $y = 2x - 5$ in the xy -plane?

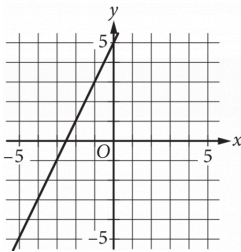
A.



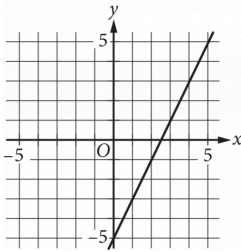
B.



C.



D.



ID: a8e6bd75 Answer

Correct Answer:

D

Rationale

Choice D is correct. In the xy -plane, the graph of the equation $y = mx + b$, where m and b are constants, is a line with slope m and y -intercept $(0, b)$. Therefore, the graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept $(0, -5)$. Having a slope of 2 means that for each increase in x by 1, the value of y increases by 2. Only the graph in choice D has a slope of 2 and crosses the y -axis at $(0, -5)$. Therefore, the graph shown in choice D must be the correct answer.

Choices A, B, and C are incorrect. The graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept at $(0, -5)$. The graph in choice A crosses the y -axis at the point $(0, 2.5)$, not $(0, -5)$, and it has a slope of $\frac{1}{2}$, not 2. The graph in choice B crosses the y -axis at $(0, -5)$; however, the slope of this line is -2 , not 2. The graph in choice C has a slope of 2; however, the graph crosses the y -axis at $(0, 5)$, not $(0, -5)$.

Question Difficulty:

Easy

Question ID 5ad6bc97

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 5ad6bc97

$$f(x) = 7x + 1$$

The function gives the total number of people on a company retreat with x managers. What is the total number of people on a company retreat with 7 managers?

ID: 5ad6bc97 Answer

Correct Answer:
50

Rationale

The correct answer is **50**. It's given that the function f gives the total number of people on a company retreat with x managers. It's also given that 7 managers are on the company retreat. Substituting 7 for x in the given function yields $f(7) = 7(7) + 1$, or $f(7) = 50$. Therefore, there are a total of **50** people on a company retreat with 7 managers.

Question Difficulty:
Easy

Question ID 1993561d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1993561d

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{4}$ and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = \frac{1}{4}x + 5$
- B. $f(x) = \frac{1}{4}x + \frac{1}{5}$
- C. $f(x) = \frac{1}{4}x - \frac{5}{4}$
- D. $f(x) = \frac{1}{4}x - 5$

ID: 1993561d Answer

Correct Answer:

A

Rationale

Choice A is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{4}$. Therefore, $m = \frac{1}{4}$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting $\frac{1}{4}$ for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = \frac{1}{4}x + 5$.

Choice B is incorrect. This equation defines a function whose graph has a y -intercept of $(0, \frac{1}{5})$, not $(0, 5)$.

Choice C is incorrect. This equation defines a function whose graph has a y -intercept of $(0, -\frac{5}{4})$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a y -intercept of $(0, -5)$, not $(0, 5)$.

Question Difficulty:

Easy

Question ID a73a5c22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a73a5c22

The function g is defined by $g(x) = 10x + 8$. What is the value of $g(x)$ when $x = 8$?

- A. 0
- B. 8
- C. 10
- D. 88

ID: a73a5c22 Answer

Correct Answer:

D

Rationale

Choice D is correct. The value of $g(x)$ when $x = 8$ can be found by substituting 8 for x in the given equation $g(x) = 10x + 8$. This yields $g(8) = 10(8) + 8$, or $g(8) = 88$. Therefore, when $x = 8$, the value of $g(x)$ is 88.

Choice A is incorrect. This is the value of x when $g(x) = 8$, rather than the value of $g(x)$ when $x = 8$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 5907e072

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 5907e072

$$f(x) = x + b$$

For the linear function f , b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b ?

- A. -30
- B. $-\frac{1}{30}$
- C. $\frac{1}{30}$
- D. 30

ID: 5907e072 Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that when $x = 0$, $f(x) = 30$. Substituting 0 for x and 30 for $f(x)$ in the given function yields $30 = 0 + b$, or $30 = b$. Therefore, the value of b is 30 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID a4d6fbec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a4d6fbec

If $y = 5x + 10$, what is the value of y when $x = 8$?

ID: a4d6fbec Answer

Correct Answer:
50

Rationale

The correct answer is 50. Substituting 8 for x in the given equation yields $y = 5(8) + 10$, or $y = 50$. Therefore, the value of y is 50 when $x = 8$.

Question Difficulty:
Easy

Question ID 8643d906

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 8643d906

$P(t) = 250 + 10t$

The population of snow leopards in a certain area can be modeled by the function P defined above, where $P(t)$ is the population t years after 1990. Of the following, which is the best interpretation of the equation $P(30) = 550$?

- A. The snow leopard population in this area is predicted to be 30 in the year 2020.
- B. The snow leopard population in this area is predicted to be 30 in the year 2030.
- C. The snow leopard population in this area is predicted to be 550 in the year 2020.
- D. The snow leopard population in this area is predicted to be 550 in the year 2030.

ID: 8643d906 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that $P(t)$ represents the population of snow leopards t years after 1990. $P(30) = 550$ corresponds to $t = 30$ and $P(t) = 550$. It follows that $t = 30$ corresponds to 30 years after 1990, or 2020. Thus, the best interpretation of $P(30) = 550$ is that the snow leopard population in this area is predicted to be 550 in the year 2020.

Choices A and B are incorrect and may result from reversing the interpretations of t and $P(t)$. Choice D is incorrect and may result from determining that 30 years after 1990 is 2030, not 2020.

Question Difficulty:
Easy

Question ID 1ecaa9c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1ecaa9c0

Robert rented a truck to transport materials he purchased from a hardware store. He was charged an initial fee of \$20.00 plus an additional \$0.70 per mile driven. If the truck was driven 38 miles, what was the total amount Robert was charged?

- A. \$46.60
- B. \$52.90
- C. \$66.90
- D. \$86.50

ID: 1ecaa9c0 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that Robert was charged an initial fee of \$20.00 to rent the truck plus an additional \$0.70 per mile driven. Let m represent the number of miles the truck was driven. Since the rental charge is \$0.70 per mile driven, $0.70m$ represents the amount Robert was charged for m miles driven. Let c equal the total amount, in dollars, Robert was charged to rent the truck. The total amount can be represented by the equation $c = 20.00 + 0.70m$. It's given that the truck was driven 38 miles, thus $m = 38$. Substituting 38 into the equation gives $c = 20.00 + 0.70(38)$. Multiplying $0.70(38)$ gives $c = 20.00 + 26.60$. Adding these values gives $c = 46.60$, so the total amount Robert was charged is \$46.60.

Choices B, C, and D are incorrect and may result from setting up the equation incorrectly or from making calculation errors.

Question Difficulty:

Easy

Question ID a232aa96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a232aa96

The function f is defined by $f(x) = -3x + 60$. What is the value of $f(x)$ when $x = -8$?

- A. 49
- B. 52
- C. 57
- D. 84

ID: a232aa96 Answer

Correct Answer:

D

Rationale

Choice D is correct. The value of $f(x)$ when $x = -8$ can be found by substituting -8 for x in the given function. This yields $f(-8) = -3(-8) + 60$, or $f(-8) = 84$. Therefore, when $x = -8$, the value of $f(x)$ is 84.

Choice A is incorrect. This is the value of $(-3 + (-8)) + 60$, not $-3(-8) + 60$.

Choice B is incorrect. This is the value of $-8 + 60$, not $-3(-8) + 60$.

Choice C is incorrect. This is the value of $-3 + 60$, not $-3(-8) + 60$.

Question Difficulty:

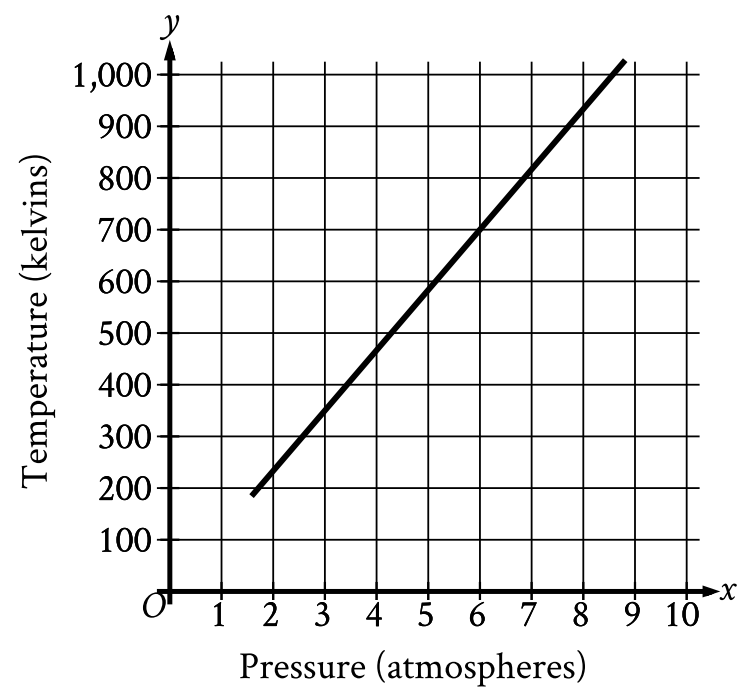
Easy

Question ID 0ea7ef01

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0ea7ef01

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y , in kelvins, of the oxygen gas when its pressure is x atmospheres.



What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres?

- A. **6**
- B. **60**
- C. **700**
- D. **760**

ID: 0ea7ef01 Answer

Correct Answer:
C

Rationale

Choice C is correct. For the graph shown, the x -axis represents pressure, in atmospheres, and the y -axis represents temperature, in kelvins. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres is represented by the y -coordinate of the point on the graph that has an x -coordinate of **6**. The point on the graph with an x -coordinate of **6** has a y -coordinate of approximately **700**. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres is **700**.

Choice A is incorrect. This is the pressure, in atmospheres, not the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 2eef7e61

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2eef7e61

The graph of the function f is a line in the xy -plane. If the line has slope $\frac{3}{4}$ and $f(0) = 3$, which of the following defines f ?

- A. $f(x) = \frac{3}{4}x - 3$
- B. $f(x) = \frac{3}{4}x + 3$
- C. $f(x) = 4x - 3$
- D. $f(x) = 4x + 3$

ID: 2eef7e61 Answer

Correct Answer:
B

Rationale

Choice B is correct. The equation for the function f in the xy -plane can be represented by $f(x) = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept. Since it's given that the line has a slope of $\frac{3}{4}$, it follows that $m = \frac{3}{4}$ in $f(x) = mx + b$, which yields $y = \frac{3}{4}x + b$. It's given that $f(0) = 3$. This implies that the graph of the function f in the xy -plane passes through the point $(0,3)$. Thus, the y -coordinate of the y -intercept of the graph is 3, so $b = 3$ in $f(x) = \frac{3}{4}x + b$, which yields $f(x) = \frac{3}{4}x + 3$. Therefore, the equation $f(x) = \frac{3}{4}x + 3$ defines the function f .

Choice A is incorrect and may result from a sign error for the y -intercept. Choice C is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$, in addition to a sign error for the y -intercept. Choice D is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$.

Question Difficulty:
Easy

Question ID 97eab129

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 97eab129

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area x , in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define f ?

- A. $f(x) = 0.6x$
- B. $f(x) = 1.8x$
- C. $f(x) = 2,268x$
- D. $f(x) = 4,536x$

ID: 97eab129 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the table represents the relationship between the area x , in square feet, of the roof of a shopping center and the amount of water $f(x)$, in gallons, drained from the roof into the gutters. Every choice represents this relationship with an equation defining f in the form $f(x) = mx$, where m is a constant rate of change. The value of m can be determined by dividing both sides of the equation by x . Each of three pairs of x and $f(x)$ in the table yield $m = 1.8$, since $\frac{4,536}{2,520} = 1.8$, $\frac{6,804}{3,780} = 1.8$, and $\frac{9,072}{5,040} = 1.8$. Therefore, the equation $f(x) = 1.8x$ could define f .

Choice A is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $0.6(2,520)$, or 1,512, gallons, not the 4,536 gallons shown in the table.

Choice C is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $2,268(2,520)$, or 5,715,360, gallons, not the 4,536 gallons shown in the table.

Choice D is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $4,536(2,520)$, or 11,430,720, gallons, not the 4,536 gallons shown in the table.

Question Difficulty:
Easy

Question ID 81390d6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 81390d6c

The function h is defined by $h(x) = x + 200$. What is the value of $h(50)$?

- A. 200
- B. 250
- C. 10,000
- D. 50,200

ID: 81390d6c Answer

Correct Answer:
B

Rationale

Choice B is correct. Substituting **50** for x in the given function yields $h(50) = 50 + 200$, or $h(50) = 250$. Therefore, the value of $h(50)$ is **250**.

Choice A is incorrect. This is the value of $h(0)$.

Choice C is incorrect. This is the value of $h(9,800)$.

Choice D is incorrect. This is the value of $h(50,000)$.

Question Difficulty:
Easy

Question ID 73b5f330

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 73b5f330

The function f is defined by $f(x) = 5x + 8$. For what value of x does $f(x) = 58$?

- A. 10
- B. 13
- C. 50
- D. 298

ID: 73b5f330 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that the function f is defined by $f(x) = 5x + 8$. Substituting 58 for $f(x)$ in this equation yields $58 = 5x + 8$. Subtracting 8 from both sides of this equation yields $50 = 5x$. Dividing both sides of this equation by 5 yields $10 = x$. Therefore, the value of x when $f(x) = 58$ is 10 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f(58)$, not the value of x when $f(x) = 58$.

Question Difficulty:

Easy

Question ID 776cfa7c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 776cfa7c

Hana deposited a fixed amount into her bank account each month. The function $f(t) = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of **25** in this context?

- A. With each monthly deposit, the amount in Hana's bank account increased by **\$25**.
- B. Before Hana made any monthly deposits, the amount in her bank account was **\$25**.
- C. After **1** monthly deposit, the amount in Hana's bank account was **\$25**.
- D. Hana made a total of **25** monthly deposits.

ID: 776cfa7c Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that t represents the number of monthly deposits. In the given function $f(t) = 100 + 25t$, the coefficient of t is **25**. This means that for every increase in the value of t by **1**, the value of $f(t)$ increases by **25**. It follows that with each monthly deposit, the amount in Hana's bank account increased by **\$25**.

Choice B is incorrect. Before Hana made any monthly deposits, the amount in her bank account was **\$100**.

Choice C is incorrect. After **1** monthly deposit, the amount in Hana's bank account was **\$125**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:

Easy

Question ID 6efcc0a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 6efcc0a3

In the linear function h , $h(0) = 41$ and $h(1) = 40$. Which equation defines h ?

- A. $h(x) = -x + 41$
- B. $h(x) = -x$
- C. $h(x) = -41x$
- D. $h(x) = -41$

ID: 6efcc0a3 Answer

Correct Answer:

A

Rationale

Choice A is correct. An equation defining a linear function can be written in the form $h(x) = ax + b$, where a and b are constants. It's given that $h(0) = 41$. Substituting 0 for x and 41 for $h(x)$ in the equation $h(x) = ax + b$ yields $41 = a(0) + b$, or $b = 41$. Substituting 41 for b in the equation $h(x) = ax + b$ yields $h(x) = ax + 41$. It's also given that $h(1) = 40$. Substituting 1 for x and 40 for $h(x)$ in the equation $h(x) = ax + 41$ yields $40 = a(1) + 41$, or $40 = a + 41$. Subtracting 41 from the left- and right-hand sides of this equation yields $-1 = a$. Substituting -1 for a in the equation $h(x) = ax + 41$ yields $h(x) = -1x + 41$, or $h(x) = -x + 41$.

Choice B is incorrect. Substituting 0 for x and 41 for $h(x)$ in this equation yields $41 = -0$, which isn't a true statement.

Choice C is incorrect. Substituting 0 for x and 41 for $h(x)$ in this equation yields $41 = -41(0)$, or $41 = 0$, which isn't a true statement.

Choice D is incorrect. Substituting 41 for $h(x)$ in this equation yields $41 = -41$, which isn't a true statement.

Question Difficulty:

Easy

Question ID fe287f7e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: fe287f7e

To repair a refrigerator, a technician charges \$60 per hour for labor plus \$120 for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

- A. $f(x) = x + 120$
- B. $f(x) = 60x$
- C. $f(x) = 60x + 120$
- D. $f(x) = 60x - 120$

ID: fe287f7e Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that the technician charges \$60 per hour for labor. Therefore, if the job takes x hours, the technician will charge $\left(\frac{\$60}{1 \text{ hour}}\right)(x \text{ hours})$, or $\$60x$, for labor. It’s also given that the technician charges \$120 for parts. Therefore, $f(x) = 60x + 120$ represents the total amount, in dollars, the technician will charge for this job if it takes x hours.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This function represents the total amount, in dollars, the technician charges for labor only, not the total amount charged for labor and parts.

Choice D is incorrect. This function represents the total amount, in dollars, the technician would charge if the charge for parts were subtracted from, rather than added to, the charge for labor.

Question Difficulty:
Easy

Question ID aad7e1b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: aad7e1b9

The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. $(-2, 0)$
- B. $(0, -2)$
- C. $(0, \frac{1}{10})$
- D. $(\frac{1}{10}, 0)$

ID: aad7e1b9 Answer

Correct Answer:

B

Rationale

Choice B is correct. The y -intercept of the graph of a function in the xy -plane is the point on the graph where $x = 0$. It's given that $f(x) = \frac{1}{10}x - 2$. Substituting 0 for x in this equation yields $f(0) = \frac{1}{10}(0) - 2$, or $f(0) = -2$. Since it's given that $y = f(x)$, it follows that $y = -2$ when $x = 0$. Therefore, the y -intercept of the graph of $y = f(x)$ in the xy -plane is $(0, -2)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID cee5b352

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: cee5b352

The length, y , of a white whale was **162 centimeters (cm)** when it was born and increased an average of **4.8 cm** per month for the first **12** months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in **cm**, of the whale?

- A. $y = 162x$
- B. $y = 162x + 162$
- C. $y = 4.8x + 4.8$
- D. $y = 4.8x + 162$

ID: cee5b352 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the length of the whale was **162 cm** when it was born and that its length increased an average of **4.8 cm** per month for the first **12** months after it was born. Since x represents the number of months after the whale was born, the total increase in the whale's length, in **cm**, is **4.8** times x , or **4.8x**. The length of the whale y , in **cm**, can be found by adding the whale's length at birth, **162 cm**, to the total increase in length, **4.8x cm**. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

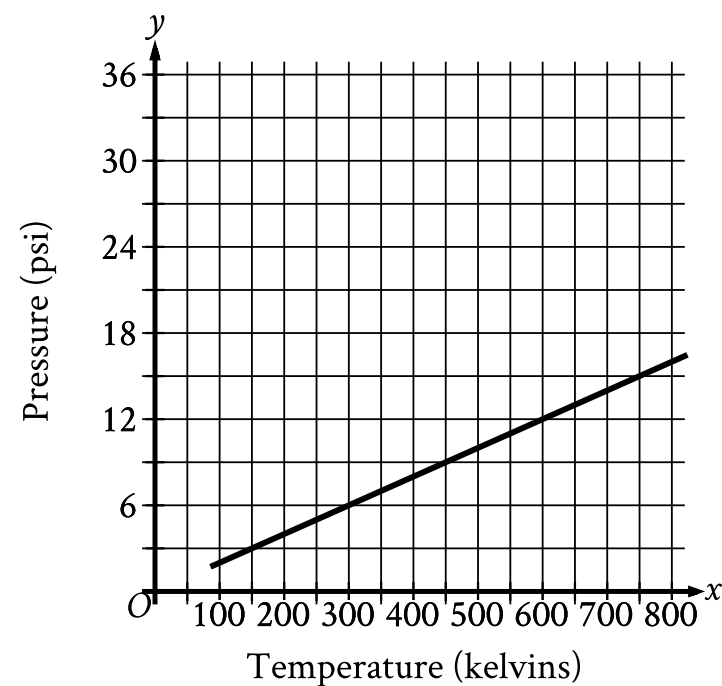
Question Difficulty:
Easy

Question ID d0cb49e8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: d0cb49e8

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure y , in **pounds per square inch (psi)**, of the argon when its temperature is x kelvins.



What is the estimated pressure of the argon, in **psi**, when the temperature is **600** kelvins?

- A. 6
- B. 12
- C. 300
- D. 600

ID: d0cb49e8 Answer

Correct Answer:
B

Rationale

Choice B is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents the estimated pressure, in **pounds per square inch (psi)**. The estimated pressure of the argon when the temperature is **600** kelvins can be found by locating the point on the graph where the value of x is equal to **600**. The graph passes through the point **(600, 12)**. This means that when the temperature is **600** kelvins, the estimated pressure is **12 psi**.

Choice A is incorrect. This is the estimated pressure, in **psi**, of the argon when the temperature is **300** kelvins, not **600** kelvins.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the temperature, in kelvins, of the argon.

Question Difficulty:

Easy

Question ID 4f89e4d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4f89e4d4

A student council group is selling school posters for a fundraiser. They use the function $p(x) = 5x - 220$ to determine their profit $p(x)$, in dollars, for selling x school posters. In order to earn a profit of \$900, how many school posters must they sell?

ID: 4f89e4d4 Answer

Correct Answer:

224

Rationale

The correct answer is **224**. It's given that a student council group uses the function $p(x) = 5x - 220$ to determine their profit $p(x)$, in dollars, for selling x school posters. Substituting **900** for $p(x)$ in the given function yields $900 = 5x - 220$. Adding **220** to each side of this equation yields $1,120 = 5x$. Dividing each side of this equation by **5** yields $224 = x$. Therefore, in order to earn a profit of \$900, they must sell **224** school posters.

Question Difficulty:

Easy

Question ID de6fe450

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: de6fe450

On January 1, 2015, a city’s minimum hourly wage was \$9.25. It will increase by \$0.50 on the first day of the year for the next 5 years. Which of the following functions best models the minimum hourly wage, in dollars, x years after January 1, 2015, where $x = 1, 2, 3, 4, 5$?

- A. $f(x) = 9.25 - 0.50x$
- B. $f(x) = 9.25x - 0.50$
- C. $f(x) = 9.25 + 0.50x$
- D. $f(x) = 9.25x + 0.50$

ID: de6fe450 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that the city’s minimum hourly wage will increase by \$0.50 on the first day of the year for the 5 years after January 1, 2015. Therefore, the total increase, in dollars, in the minimum hourly wage x years after January 1, 2015, is represented by $0.50x$. Since the minimum hourly wage on January 1, 2015, was \$9.25, it follows that the minimum hourly wage, in dollars, x years after January 1, 2015, is represented by $9.25 + 0.50x$. Therefore, the function $f(x) = 9.25 + 0.50x$ best models this situation.

Choices A, B, and D are incorrect. In choice A, the function models a situation where the minimum hourly wage is \$9.25 on January 1, 2015, but decreases by \$0.50 on the first day of the year for the next 5 years. The functions in choices B and D both model a situation where the minimum hourly wage is increasing by \$9.25 on the first day of the year for the 5 years after January 1, 2015.

Question Difficulty:
Easy

Question ID 13909d78

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 13909d78

The function f is defined by the equation $f(x) = 100x + 2$. What is the value of $f(x)$ when $x = 9$?

- A. 111
- B. 118
- C. 900
- D. 902

ID: 13909d78 Answer

Correct Answer:
D

Rationale

Choice D is correct. Substituting 9 for x in the given equation yields $f(9) = 100(9) + 2$, or $f(9) = 902$. Therefore, the value of $f(x)$ when $x = 9$ is 902.

Choice A is incorrect. This is the value of $f(x)$ when $x = 1.09$.

Choice B is incorrect. This is the value of $f(x)$ when $x = 1.16$.

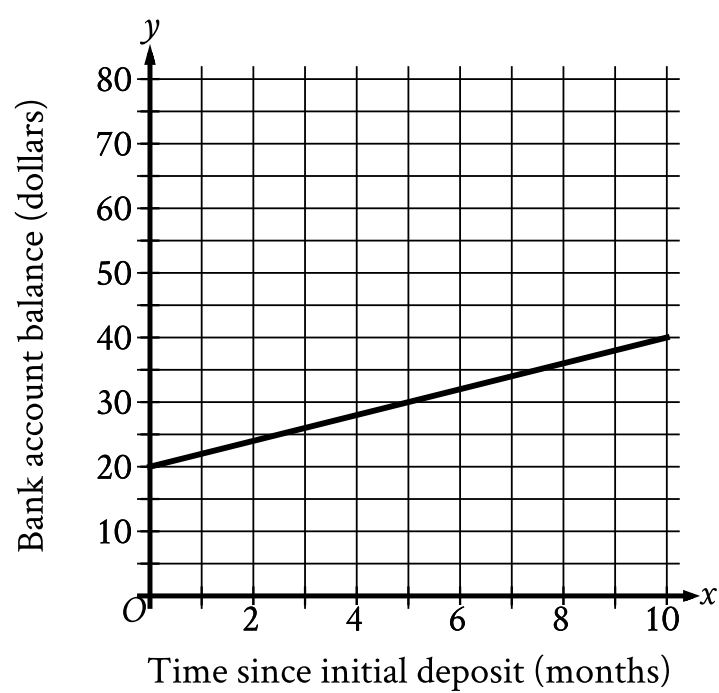
Choice C is incorrect. This is the value of $f(x)$ when $x = 8.98$.

Question Difficulty:
Easy

Question ID 2ecce641

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2ecce641



A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

ID: 2ecce641 Answer

Correct Answer:
20

Rationale

The correct answer is **20**. For the graph shown, the x -axis represents the time since the initial deposit, in months, and the y -axis represents the bank account balance, in dollars. The amount of the initial deposit is estimated by the y -coordinate of the point on the graph that represents **0** months since the initial deposit. Therefore, the amount of the initial deposit is estimated by the corresponding y -value for the point when $x = 0$. When $x = 0$, it is estimated that $y = 20$. Thus, the amount of the initial deposit estimated by the graph, to the nearest whole dollar, is **20**.

Question Difficulty:
Easy

Question ID a1696f3e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a1696f3e

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

- A. 30
- B. 22
- C. 11
- D. -23

ID: a1696f3e Answer

Correct Answer:
C

Rationale

Choice C is correct. Substituting 4 for x in $g(x) = 5x + a$ gives $g(4) = 5(4) + a$. Since $g(4) = 31$, the equation $g(4) = 5(4) + a$ simplifies to $31 = 20 + a$. It follows that $a = 11$.

Choices A, B, and D are incorrect and may result from arithmetic errors.

Question Difficulty:
Easy